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A Text Generation Hallucination Detection Framework Based on Fact and Semantic Consistency

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***Abstract*-With the development of pre-trained language models, significant progress has been made in text generation. However, these models often face inconsistencies with facts and contradictions in sentences, known as the "hallucination" problem of language models. To address this issue, HalluCheckNet, a mechanism thatintegrates factual and semantic consistency for hallucination detection, is proposed. By combining pre-trained models with the Plug-and-Play Language Model (PPLM) framework, HalluCheckNet treats the degree of hallucination as a controllable attribute, using controlled text generation techniques to effectively reduce model hallucinations. Experiments conducted on GPT2 have achieved high-quality generation of social media, review, and news texts. The results demonstrate that this method not only maintains text fluency but also significantly reduces factual and semantic errors, making the generated texts more coherent and credible.**

Keywords-Text Generation Language Model Hallucination Semantic Consistency, Factual Consistency, PPLM

I. INTRODUCTION

In contemporary natural language processing research, text generation models like ChatGPT have achieved remarkable success in producing naturally fluent texts. However, these models tend to generate content that is either factually inconsistent [1] or logically disordered [2], referred to as "hallucinations," especially in complex or data-scarce domains. This study introduces a text generation hallucination detection framework based on factual and semantic consistency to identify and correct hallucinations in text generation. Systematic experiments were designed to validate the model, laying a foundation for subsequent text analysis. By applying this mechanism to general text generation, we enhanced the factual accuracy and logical consistency of the generated text, offering a more robust and trustworthy solution for natural language generation.

II. THEORIES RELATED TO TEXT GENERATION

A. Language Modeling

In the process of language modeling, the goal is to master the probability distribution of word sequences. Models are typically trained using Maximum Likelihood Estimation (MLE) with the following formula:

$$LMLE = -\sum_{t=1}^{T} \log p\,(x_t|x_{<t})$$
$$p_\theta(x) = \prod_{t=1}^{T+1} p_\theta(x_t|x_{<t}) \quad (1)$$

B. GPT-2 Open Text Generation

GPT-2[3] employs an autoregressive training approach, utilizing vast amounts of web data for unsupervised training and adapting to specific downstream tasks through fine-tuning. It features multiple levels of the Transformer Decoder architecture, processing the input text's context with a multi-head self-attention mechanism, followed by a feed-forward network layer to generate the output probability distribution of the target word. The computation formula is as follows:

$$h_z = transformer_block(h_{z-1}) \forall z \in [1, n]$$
$$p(c) = softmax(h_z w_e^T) \quad (2)$$

C. PPLM Conditional Text Generation

The PPLM framework [4] introduces a method to address the challenges of control and scalability in text generation. It combines one or several classifier models, designed for specific attributes like mood or theme, with a pre-trained unconditional language model. Through the classifier model, the attributes of the generated text are classified and predicted, and these predictions are used to adjust the internal parameters of the language model. This process aims to increase the probability of generating text that aligns with the desired attributes. The updated parameters are then used to generate new vocabulary, with the parameter updating process repeated at each time step to guide the text generation more accurately towards the desired attributes. The computational formula is as follows:

$$p(x|a) \propto p(a|x)p(x)$$
$$(x_{t+1}, H_{t+1}) = LM(x_t, H_t) \quad (3)$$

Where x_{t+1} is the text generated through softmax sampling, H_t is the hidden layer information at step t, and the historical matrix H_t is updated through ΔH_t, allowing for the re-prediction of x_{t+1}:

$$\Delta H_t \leftarrow \Delta H_t + \alpha \nabla \Delta Ht \log p\,(a/H_t + \Delta H_t) \nabla \Delta Ht \log p\,(a/H_t + \Delta H_t) \gamma \quad (4)$$

Where, α is the step size, and γ is the ratio of the normalization term.

III. HALLUCINATION DETECTION MECHANISM

This section introduces in detail the proposed hallucination detection mechanism, HalluCheckNet, which assesses and corrects generated text from both semantic consistency and factual

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accuracy perspectives. Figure 1 illustrates the overall structure of HalluCheckNet.

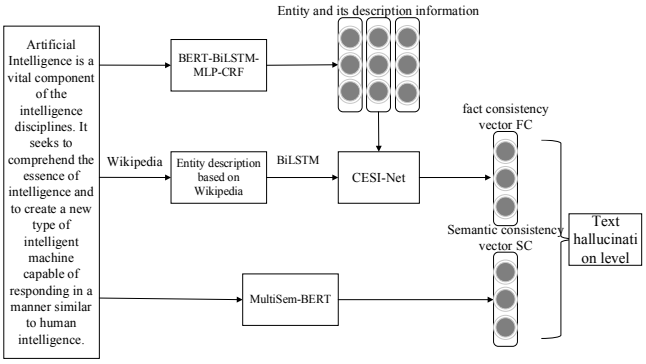


Fig. 1. Overall Structure of HalluCheckNet

In Figure 1, for a given text, the entity recognition model BERT-BiLSTM-MLP-CRF proposed in this paper is used to identify entities and their descriptions. Then, the open-source entity annotation tool HanLp by He et al [5]. is utilized to extract entities from the text. For each entity identified by HanLP, the Wikipedia API is employed to search for the corresponding Wikipedia page, from which content is retrieved and parsed to obtain entity descriptions based on Wikipedia. These two types of entity representations are then inputted into the Chinese Entity Comparison Network to calculate the Factual Consistency (FC) vector. Additionally, the text is fed into the BERT-based multi-task learning model, MultiSem-Bert, proposed in this paper to obtain the Semantic Consistency (SC) vector. Finally, these two parts serve as a judging model to score the generated text from both factual consistency and semantic consistency perspectives, guiding text generation through gradient backpropagation within the PPLM framework, thereby reducing the occurrence of hallucinations during text generation.

A. Factual Consistency Module

1) Overview of the BERT-BiLSTM-MLP-CRF Model

The BERT-BiLSTM-MLP-CRF model proposed in this paper integrates multiple layers to enhance the accuracy of identifying entities and their descriptions in Chinese texts. The model's structure is designed as follows (figure 2):

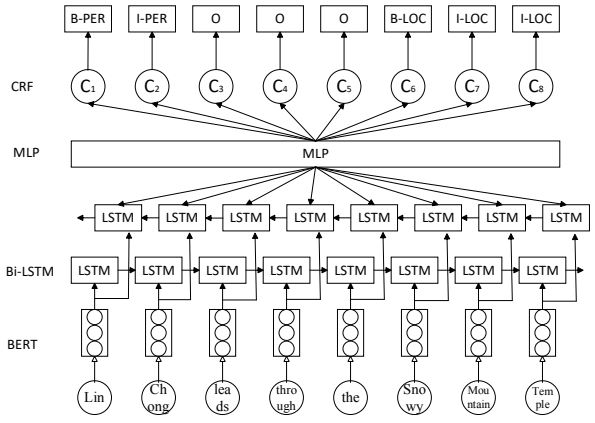


Fig. 2. Structure of the BERT-BiLSTM-MLP-CRF Model

(1) BERT Layer: The BERT layer captures deep semantic features of each word in the text through a pre-trained deep bi-directional language model. For input text $x_1, x_2, \ldots, x_n$, the BERT layer encodes it into hidden states $h_1, h_2, \ldots, h_n$, where $h_i = BERT(x_i)$.

(2) BiLSTM Layer: The BiLSTM layer, with its forward and backward LSTM networks, accurately captures the temporal association between entities and their descriptions in lengthy texts. This layer takes the output from the BERT layer $h_1, h_2, \ldots, h_{n_k}$ and outputs hidden vectors $h'_1, h'_2, \ldots, h'_n$, where $h'_{i1} = \mathrm{LSTM}_{\mathrm{forward}}(h_i)$ and $h'_{i2} = \mathrm{LSTM}_{\mathrm{backward}}(h_i)$, with the final $h'_i = h'_{i1} \oplus h'_{i2}$.

(3) MLP Layer: The MLP layer integrates and weights entities and their descriptions through a multi-layer network structure, taking the output of the BiLSTM layer as input and outputting integrated features: $\mathrm{c}_1, \mathrm{c}_2, \ldots, \mathrm{c}_n$, with $\mathrm{c_i} = \mathrm{MLP}(h'_\mathrm{i})$.

(4) CRF Layer: The CRF layer improves the accuracy of entity recognition by optimizing the final tagging sequence, considering the dependency between adjacent tags. It takes the output of the MLP layer $\mathrm{c}_1, \mathrm{c}_2, \ldots, \mathrm{c}_\mathrm{n}$ and generates the final tagging sequence $\mathrm{y}_1, \mathrm{y}_2, \ldots, \mathrm{y}_\mathrm{n}$.The computation formula is as follows:

$$p(y|x) = \frac{exp\left(\sum_{i=1}^{n} W_{ii}, y_{i-1} + b_{yi}\right)}{\sum_{i=1}^{n} exp\left(\sum_{i=1}^{n} W'_{i,ji}, \ldots + b'_{ij}\right)} \quad (5)$$

Where W and b are the CRF layer's weights and biases respectively.

The BERT-BiLSTM-MLP-CRF model employs a multi-task learning loss function composed of two main parts:

(1) Entity Recognition Loss: Generated by the CRF layer to maximize the probability of the correct label sequence, using the negative log likelihood loss from the CRF layer's output, expressed as

$$L_{entity} = -\sum_{i=1}^{N} log\ (p(y_i \mid x_i)) \quad (6)$$

Where N is the number of training samples, with $\mathrm{x_i}$ and $\mathrm{y_i}$ representing the model's input and label sequence respectively.

(2) Description Information Recognition Loss: Utilizes Binary Cross-Entropy (BCE) for recognizing descriptive information, treating it as a classification task for each possible descriptive information label, with the loss function expressed as

$$L_{desc} = -\frac{1}{M}\sum_{j=1}^{M} [y'_j log\ (\sigma(x'_j)) + (1 - y'_j) log\ (1 - \sigma(x'_j))] \quad (7)$$

where M is the number of possible descriptive information labels, $\mathrm{x'_j}$ is the model's predictive output for the jth descriptive information label, $\mathrm{y'_j}$ is the corresponding true label, and σ is the sigmoid function.

The total model loss is a weighted sum of these two parts.

2) Overview of the Chinese Entity Comparison Network, CESI-Net

CESI-Net (Chinese Entity Similarity Inference Network) is a network specifically designed for Chinese texts. It evaluates

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whether the descriptions of entities in the generated text are consistent with those in Wikipedia by calculating similarity scores and inference scores between entities and their descriptions in the text and corresponding entity descriptions in Wikipedia. The structure of CESI-Net is illustrated in Figure 3.

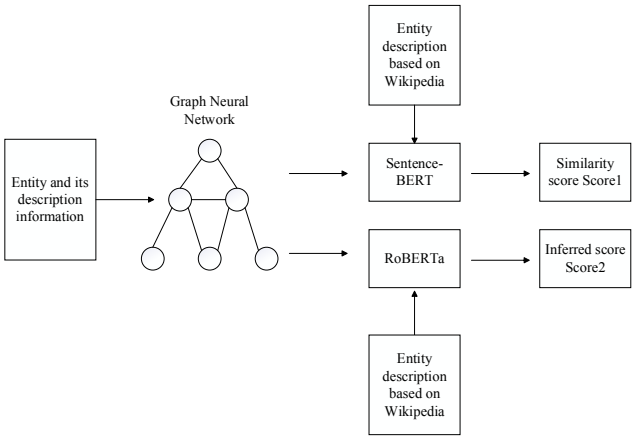


Fig. 3. Structure of CESI-Net

Specifically, CESI-Net utilizes the previously mentioned BERT-BiLSTM-MLP-CRF model to extract entities and their descriptions from the text. These extracted entities and descriptions are then fed into a Graph Neural Network (GNN) to model the complex relationships between entities. The GNN updates node features to learn the semantic associations between entities, thus producing a comprehensive entity relationship graph. Subsequently, the output from the GNN is inputted into Sentence-BERT [6] and RoBERTa [7] models. Sentence-BERT is used to calculate the similarity score (Score1) between the entity descriptions in the text and those in Wikipedia. The RoBERTa model assesses the inference score (Score2) between the entity descriptions in the text and Wikipedia. The final Score is calculated by combining the similarity score, Score1, and inference score, Score2, through a weighted average.

B. Semantic Consistency Module

1) Overview of the MultiSem-BERT Model

The MultiSem-BERT model proposed in this paper is a BERT-based multi-task learning model that integrates a dual attention mechanism, specifically designed for natural language inference tasks. It encompasses two main tasks: contradiction detection and thematic consistency detection. The structure of the model is depicted in Figure 4.

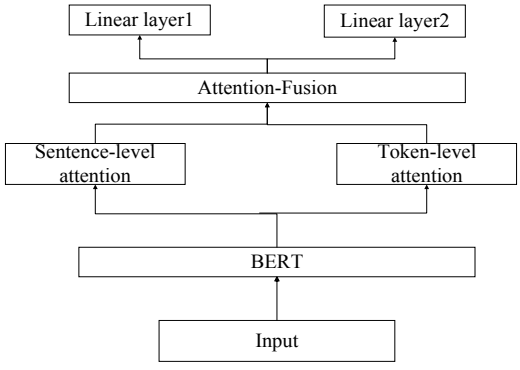


Fig. 4. Structure of the MultiSem-BERT Model

(1) Token-Level Attention Mechanism: The token-level attention mechanism focuses on analyzing the relationship between each word in a sentence pair and all words in the other sentence. For each word, the model generates a weighted token-level representation by calculating its relationship weights with all words in the other sentence.

(2) Sentence-Level Attention Mechanism: The sentence-level attention mechanism concentrates on analyzing the global semantic relationship between sentences. For each sentence, it first performs word segmentation using the Jieba segmentation library, and then assigns different weights based on part of speech (such as nouns, verbs, and conjunctions). Based on these weighted parts of speech, the model calculates the relationship weight between sentences to obtain a sentence-level weighted representation.

The MultiSem-BERT model utilizes a weighted loss function from multitask learning to simultaneously optimize contradiction detection and topic consistency detection tasks. Both the contradiction detection task and the topic consistency detection task use the binary cross-entropy loss function, with the calculation formula being the same as Equation 7. The total loss function of the model is a weighted sum of the losses from both tasks, with the weights being learnable parameters, calculated as follows:

IV. Experimental Setup and Results Analysis

A. Open-Ended Text Generation Task

In the open-ended text generation task, we used the Weibo dataset, comments dataset, and news dataset, which were made public by Liang et al. Special symbols and redundant tags were removed from the datasets through regular expressions, resulting in three high-quality datasets, each containing 3 million entries. These datasets were then used to train the GPT2-Chinese model, producing three fine-tuned models: GPT2-Chinese-Weibo, GPT2-Chinese-Review, and GPT2-Chinese-News. The experiment was deployed on the cloud computing platform AutoDL and developed using Python.

B. Hallucination Detection Mechanism

1) Fact Consistency Module

To verify the capability of the BERT-BiLSTM-MLP-CRF model in identifying entities and their descriptive information, the experiment compared the performance of various models on the Named Entity Recognition (NER) dataset made public by Liang et al. The models were evaluated based on accuracy, recall, and F1 score in the task of entity and descriptive information recognition. The experimental results are presented in Table 1.

Table 1. Comparison of Model Recognition Effects (Unit: %)

Model	Acc	Recall	F1
BERT-BiLSTM-CRF	83.84	84.20	83.88
BERT-BiLSTM-MLP-CRF	84.47	85.05	84.12
BERT-BiLSTM-MLP-	86.44	87.51	86.73

The experimental results indicate that the BERT-BiLSTM-MLP-CRF model proposed in this paper outperforms other models across all evaluation metrics. Furthermore, the integration of GNN (Graph Neural Network) into the BERT-BiLSTM-MLP-

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CRF model for entity relationship modeling significantly enhances the model's performance. This improvement suggests that combining GNN can strengthen the model's understanding of the relationships between entities, thereby proving its effectiveness in tasks related to fact consistency.

2) Semantic Consistency Module

To validate the performance of the MultiSem-BERT model in discriminating contradictions and maintaining topic consistency in text, the experiment compared the improved model's performance with other models on the CMNLI dataset, which was made public by Liang et al [8]. The experimental results are shown in Table 2.

Table 2. Comparison of Model Effects (Unit: %)

Model	Discriminating contradictions			Topic consistency			
	Acc	Recall	F1	Acc	Recall	F1	avg
RoBERTa	84.06	83.45	83.70	73.67	73.21	72.98	78.85
BERT	82.98	81.89	81.75	76.60	76.47	75.96	79.27
RoBERTa+dual attention	86.87	85.88	85.70	75.87	74.45	74.32	80.52
BERT+dual attention	84.47	84.12	83.97	78.12	78.01	77.87	81.09

The experimental results demonstrate that the dual attention mechanism enhances the model's understanding of the relationships between sentences. The proposed MultiSem-BERT model exhibits excellent performance in identifying sentence contradictions and maintaining thematic consistency, proving its effectiveness in semantic consistency tasks.

C. *Effectiveness Test of the Hallucination Detection Mechanism*

To verify the effectiveness of the hallucination detection mechanism in improving text generation quality, this paper compares the performance of three fine-tuned GPT2-Chinese models before and after incorporating the hallucination detection mechanism across various automated evaluation metrics. Besides the commonly used Perplexity, this paper also adopts Fluency, Redundancy, Logical Consistency, and Factual Accuracy as evaluation metrics. The experimental results are shown in Table 3.

Table 3. Evaluation Results

	Model	Perplexity	Fluency	Redundancy	Logical Consistency	Factual Accuracy
GPT2	GPT2-Chinese-weibo	21.8	0.68	0.45	0.58	0.67
	GPT2-Chinese-Review	19.7	0.74	0.47	0.61	0.57
	GPT2-Chinese-News	17.4	0.81	0.32	0.79	0.81
GPT2+ HCN	GPT2-Chinese-weibo+HCN	20.1	0.71	0.32	0.68	0.79
	GPT2-Chinese-Review+HCN	17.9	0.77	0.35	0.78	0.78
	GPT2-Chinese-News+HCN	15.3	0.83	0.26	0.85	0.83

The experimental results show that after incorporating the hallucination detection mechanism, the models were able to improve logical consistency and factual accuracy without compromising fluency. These findings effectively demonstrate the hallucination detection mechanism's capability to enhance the model's handling of text consistency and accuracy. It also shows its generalizability across different models, thereby validating the correctness of the methods and approaches discussed in this paper.

V. CONCLUSION

This paper introduces a text generation technology based on a hallucination detection mechanism, aimed at improving the factual accuracy and logical consistency of generated texts. It proposes the BERT-BiLSTM-MLP-CRF model and CESI-Net for factual consistency, utilizing external knowledge bases to identify and correct factual errors. The MultiSem-BERT model is proposed for semantic consistency, ensuring sentence coherence by analyzing textual logical relations. The approach combines pre-trained language models with the PPLM framework for model guidance and control during generation. Despite its simplicity and efficiency, further experiments on advanced models like GPT-3 due to resource constraints are planned, aiming at exploring optimization on large models for enhanced text generation.

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